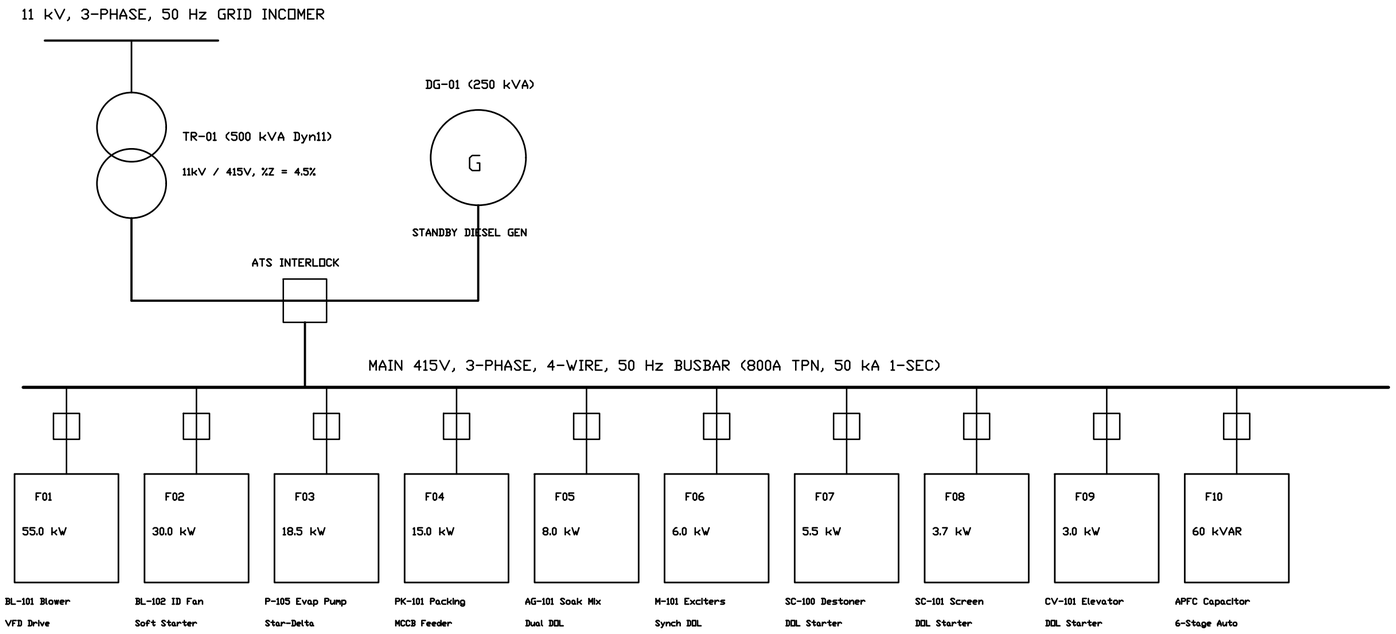
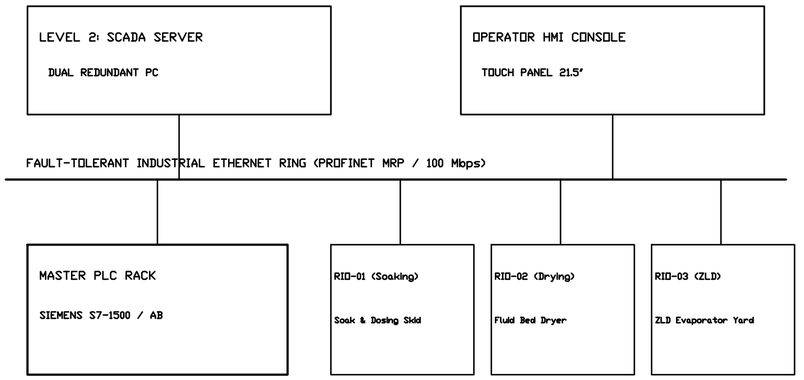


INDUSTRIAL SCADA & PLC AUTOMATION ARCHITECTURE (ISA-95 / IEC 62443):



LEVEL 0 FIELD INTERFACE: HART 4-20mA Analog Loops, Modbus RS-485 Power Meters, VFDs on PROFINET

SAFETY INTEGRITY: Dual E-Stop Safety Relays (Cat 4 / SIL 3) hardwired to MCC Main Incomer Trips

ELECTRICAL LOAD SUMMARY, MOTOR FEEDER SIZING & SWITCHGEAR SCHEDULE:

1. TOTAL CONNECTED LOAD: 184.5 kW | DIVERSITY FACTOR: 0.70 | PEAK RUNNING DEMAND: 129.2 kW (131.8 kVA @ 0.98 PF).

2. POWER FACTOR CORRECTION: 60 kVAR Automatic APFC Bank maintains bus power factor at >= 0.98 lag (Penalty free).

3. TRANSFORMER SIZING: 500 kVA, 11kV/415V, Dyn11, ONAN cooling. Loading: 26.4% (Substantial headroom for Phase 2).

4. STANDBY GENERATION: 250 kVA (200 kW @ 0.8 PF) Diesel Generator with automatic mains failure (AMF) and ATS interlock.

5. UNINTERRUPTIBLE POWER SUPPLY (UPS): 10 kVA Online industrial true-sine UPS powering PLC, SCADA, and safety instruments.

6. SHORT CIRCUIT RATING: 50 kA for 1 second fault withstand rating on all MCC switchgear, busbars, and circuit breakers.

7. EARTHING & PROTECTION: Combined TN-S system with dedicated chemical copper earthing pits (<1.0 ohm resistance).

8. LIGHTNING PROTECTION: Early Streamer Emission (ESE) air terminal per IS/IEC 62305 with 45m protection radius.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

ELECTRICAL SINGLE-LINE DIAGRAM & AUTOMATION SCADA TOPOLOGY
DISCIPLINE: ELECTRICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-011	REV: V8.0	SHEET: 11 OF 12
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ